

Research Document

Project Name: Navigators

Topic: AI/ML for Intelligent Dead Reckoning using IO-VNBD

1. Dataset Analysis: IO-VNBD

The IO-VNBD (Inertial Odometry Vehicle Navigation Benchmark Dataset) is a pivotal resource for this project.

- **Origin:** Collected across the UK, Nigeria, and France.
- **Scale:** Contains ~58 hours of smartphone-recorded data covering ~4,400 km, alongside vehicle-extracted data.
- **Challenge Addressed:** Low-cost INS sensors (like those in smartphones) suffer from substantial noise and bias, causing rapid drift when integrated twice for position.

2. State-of-the-Art ML for Dead Reckoning

- **Traditional Approach:** Pure mathematical integration of IMU data (accelerometer for velocity/position, gyroscope for heading) leads to cubic or exponential error growth.
- **AI/ML Approach (End-to-End):** Models like RoNIN (Robust Neural Inertial Navigation) or TLIO (Tight Learning-based Inertial Odometry) map sequences of IMU data directly to velocity vectors or relative displacements.
- **Sequence Models:** LSTMs and Temporal Convolutional Networks (TCNs) are highly effective because they can learn the implicit temporal patterns of human or vehicle motion, correcting for vibration and dynamic biases better than traditional filters.

3. Sensor Fusion and EKF

While AI predicts relative motion well, an absolute reference is needed to prevent unbounded drift over long periods.

- **Extended Kalman Filter (EKF):** The standard algorithm for fusing the AI's high-frequency relative predictions with low-frequency absolute GNSS updates.
- **Zero Velocity Updates (ZUPT):** A crucial heuristic where the model detects if the vehicle has stopped (e.g., at a red light) to reset accumulated velocity error to zero.

4. Map Matching

- **Purpose:** To snap the estimated trajectory to known road networks, further bounding the error.
- **Implementation:** Usually handled using Hidden Markov Models (HMM) or geometry-based snapping using offline maps (OpenStreetMap data).